

**Public Finance**

**BIEM – Spring 2026**

**Introduction to Public Finance**



# Course organisation

## Coordinator:

**Paola Profeta, full professor of Public Economics, Bocconi University**

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**class 32:** teachers Paola Profeta/Dalila Lindov

TA: Matilde Bontempo ([dalila.lindov@unibocconi.it](mailto:dalila.lindov@unibocconi.it); [matilde.bontempo@unibocconi.it](mailto:matilde.bontempo@unibocconi.it))

**class 33:** teachers Laura Bonacorsi/Andrea Riganti

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**class 34:** teacher Dalila Lindov

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# Evaluation: Written exams

You have two options.

## Option A:

First **midterm** (50%) on the first half of the course +

Second **midterm** (50%) on the second half of the course

No minimum grade is required in the first midterm exam in order to gain access to the second. The overall grade is a simple average of the grades obtained in the two midterm exams, rounded upwards.

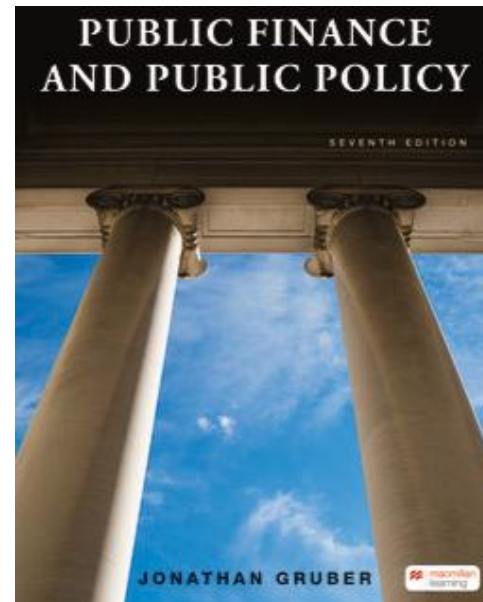
## Option B:

**General exam** (100%) on all the topics



## Course material

- Main textbook: Gruber, J., *Public Finance and Public Policy*, 7th edition, 2022
- Teaching material on blackboard (slides, exercises, additional material)



# Schedule

| Lesson                                                       | Topic                                        | Book chapter                         |
|--------------------------------------------------------------|----------------------------------------------|--------------------------------------|
| 1                                                            | Introduction to Public Finance               |                                      |
| 2                                                            | Externalities and Public Goods               | Ch 5, Ch 7, Ch 9.1                   |
| 3                                                            | Externalities and Public Goods               | Ch 5, Ch 7, Ch 9.2                   |
| 4                                                            | Education                                    | Ch 11.1, Ch 11.2, Ch 11.4, Ch 11.5   |
| 5                                                            | Education                                    | Ch 11.1, Ch 11.2, Ch 11.4, Ch 11.5   |
| 6                                                            | Social Insurance                             | Ch 12.1 - Ch 12.4                    |
| 7                                                            | Healthcare System                            | Ch 15.1 - Ch 15.4, Ch 16.1 - Ch 16.3 |
| 8                                                            | Healthcare System                            | Ch 15.1 - Ch 15.4, Ch 16.1 - Ch 16.3 |
| 9                                                            | Social Security (Pension System)             | Ch 13.1 - Ch 13.4                    |
| 10                                                           | Social Security (Pension System)             | Ch 13.1 - Ch 13.4                    |
| 11                                                           | Social Security (Pension System)             | Ch 13.1 - Ch 13.4                    |
| 12                                                           | Exam simulation                              |                                      |
| <b>FIRST MIDTERM EXAM on March 19th @ 2pm</b>                |                                              |                                      |
| 13                                                           | Introduction to Taxes                        |                                      |
| 14                                                           | Taxation                                     | Ch 18.1, Ch 18.3 - Ch 18.6           |
| 15                                                           | Taxation                                     | Ch 18.1, Ch 18.3 - Ch 18.6           |
| 16                                                           | Tax Inefficiencies                           | Ch 20                                |
| 17                                                           | Tax Inefficiencies                           | Ch 20                                |
| 18                                                           | Taxes on Savings                             | Ch 22                                |
| 19                                                           | Taxes on Savings                             | Ch 22                                |
| 20                                                           | Taxes on Risk Taking and Wealth              | Ch 23                                |
| 21                                                           | Taxes on Risk Taking and Wealth              | Ch 23                                |
| 22                                                           | Fundamental Tax Reform and Consumption Taxes | Ch 25, Ch 17.5                       |
| 23                                                           | Fundamental Tax Reform and Consumption Taxes | Ch 25, Ch 17.5                       |
| 24                                                           | Exam simulation                              |                                      |
| <b>SECOND MIDTERM EXAM or GENERAL EXAM on May 20th @ 9am</b> |                                              |                                      |



# Why Study Public Finance?

- Government plays a central role in modern economies
- Taxes and spending affect:
  - growth
  - inequality
  - incentives

**The goal of public finance is to understand the proper role of the government in the economy.**



# Why Study Public Finance Today?

Tax reform

Climate policy

Inequality

Aging populations



# Governments Are Economically Large

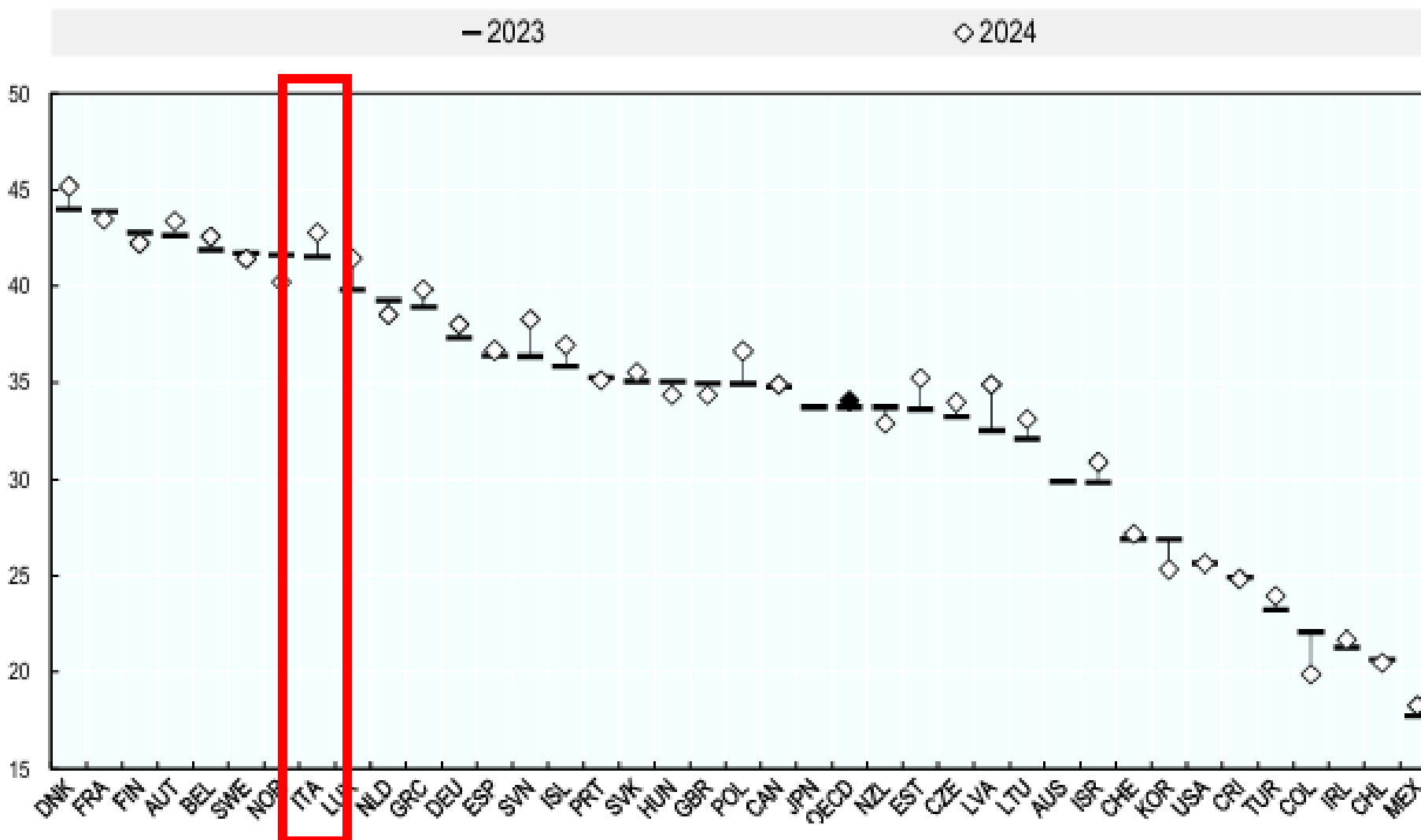
- In advanced economies:
  - public spending  $\approx$  30–55% of GDP
- Large differences across countries
- Reflect institutional and political choices

Source: OECD (2024), *Government at a Glance*





## Tax-to-GDP ratios in 2023 and 2024 (as % of GDP)



**Tax-to-GDP  $\approx$  size of government in public finance**

Source: OECD (2025), *Revenue Statistics*; Gruber

# Growth of Government Over Time

Government size has increased dramatically since 1900

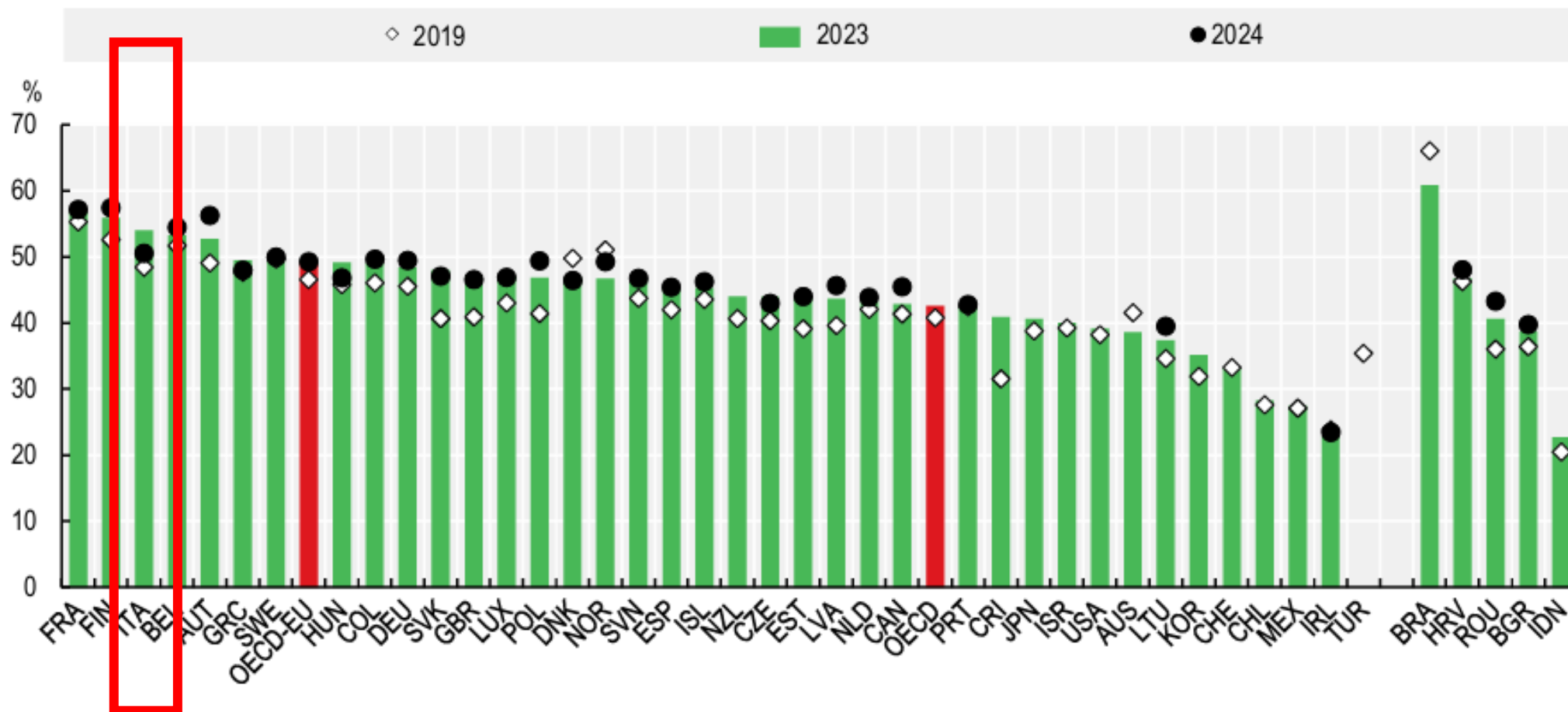
Driven by:

- social insurance
- education
- health care

Source: Gruber (Ch.1); Lindert (2004)

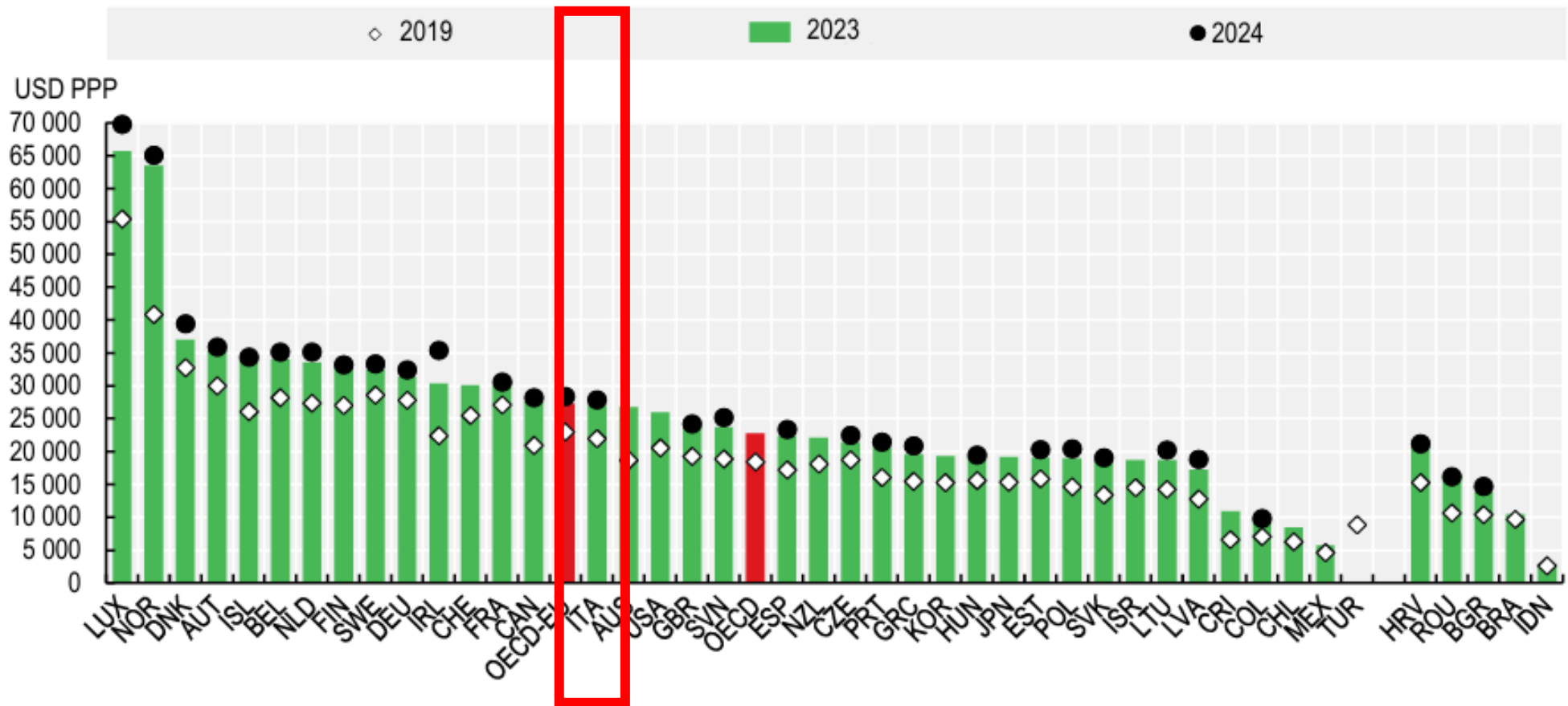


## General government expenditures as a percentage of GDP, 2019, 2023 and 2024



Source: OECD (2025), *Government at a Glance*

General government expenditures per capita, 2019, 2023 and 2024



Source: OECD (2025), *Government at a Glance*



# What Does Government Do?

Public goods  
Infrastructure, defense

Redistribution  
Taxes & transfers

Regulation  
Market rules

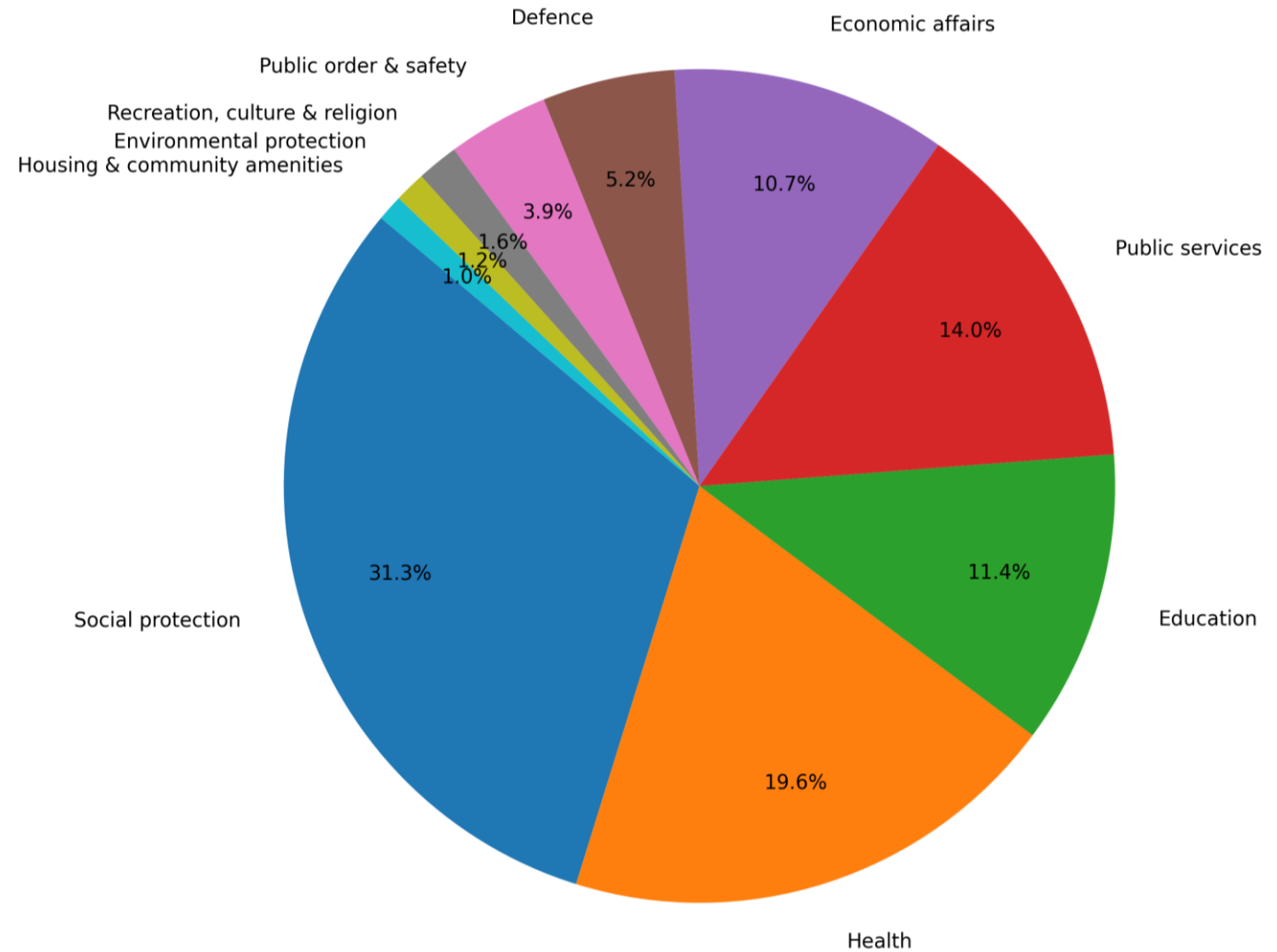
Stabilization  
Business cycles

Source: Eurostat (2023); OECD (2023)



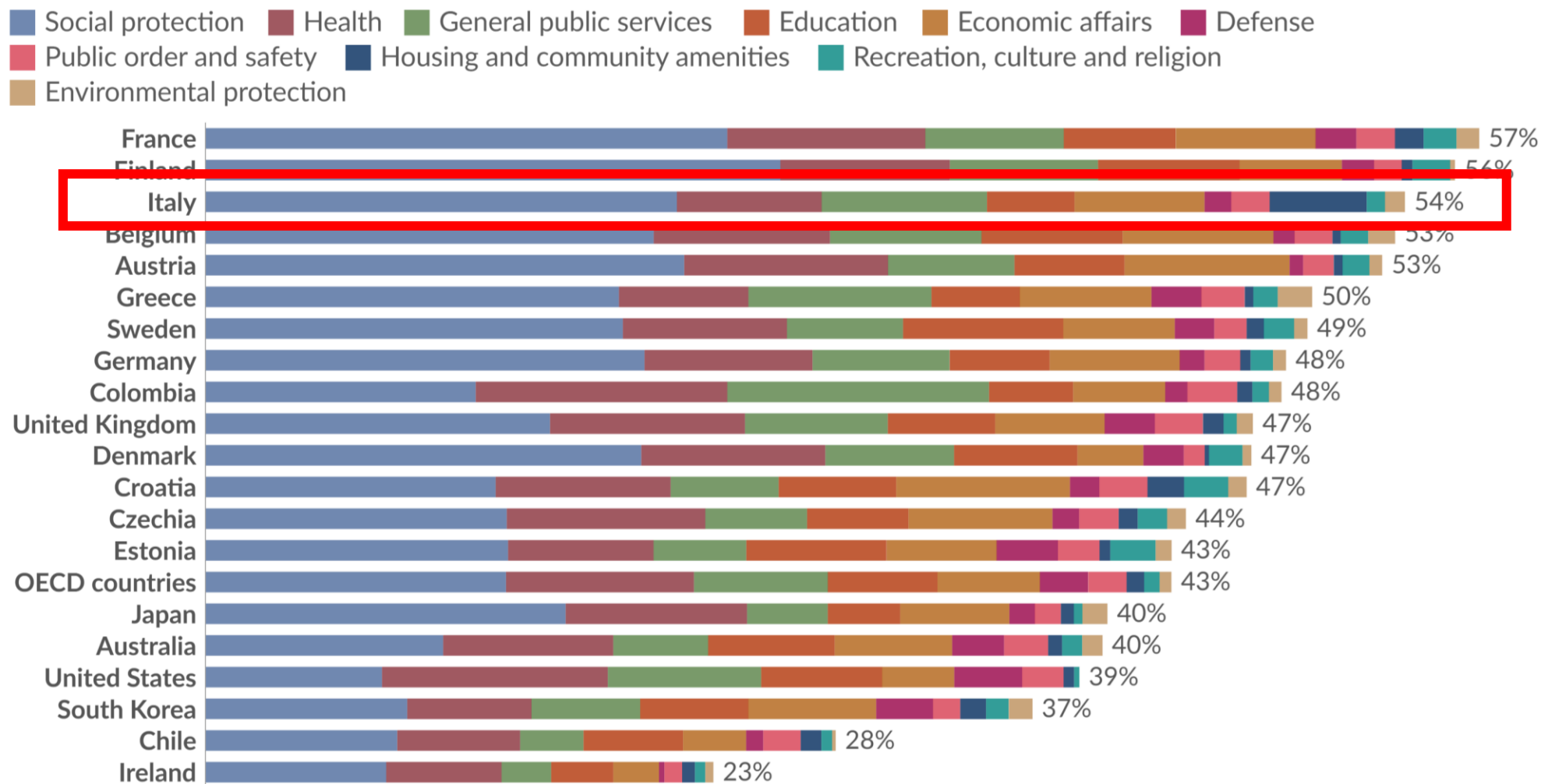
# Composition of Public Spending

Government Spending by Function (% of GDP, OECD average, 2023)



Università Commerciale  
Luigi Bocconi

## Government spending by function as a share of GDP, 2023



Source: OECD (2025); *Our World in Data*

## The Four Questions of Public Finance

1. When should the government intervene?

2. How should it intervene?

3. What are the effects of intervention?

4. Why do governments intervene the way they do?





# 1. When Should the Government Intervene?

Two reasons for the government intervention:

The market is not always able to achieve allocative *efficiency* due to the presence of market failures(e.g. public good, externality, asymmetric information)

**I Market failures:** problems that cause the market economy to deliver an outcome that does not maximize efficiency: government intervention is potentially enhancing the efficiency

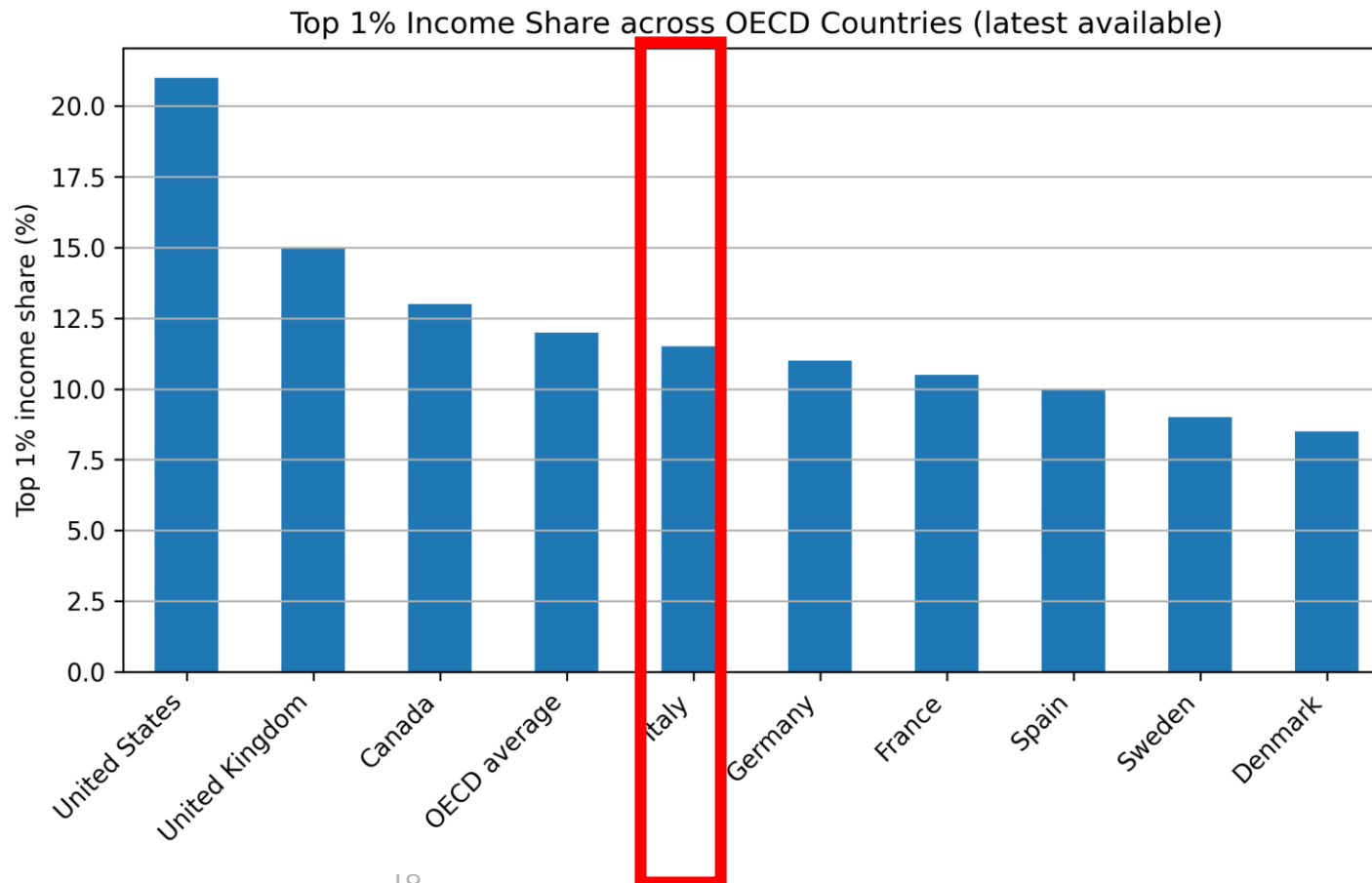
..but even if the market is well-functioning, an efficient outcome is not necessarily *socially desirable*.

**II Redistribution** is a second reason for government intervention. It represents the shifting of resources from some groups in society to others, usually motivated by some equity principle. Different forms of redistribution.



# Rising Inequality

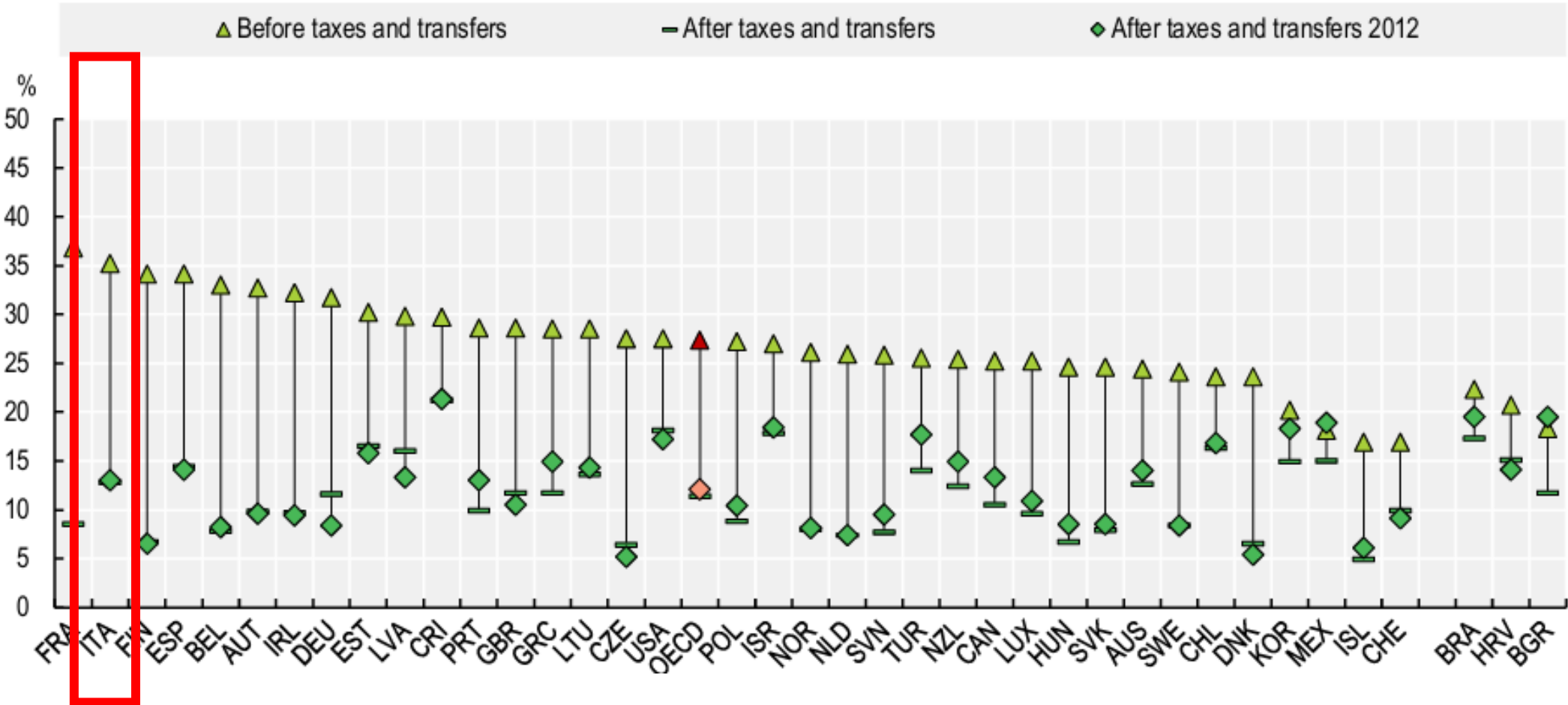
- Income inequality has increased in many countries
- Especially strong at the top of the distribution



**Source:** World Inequality Database (WID.world). See Piketty, Saez & Zucman (2018, 2019).

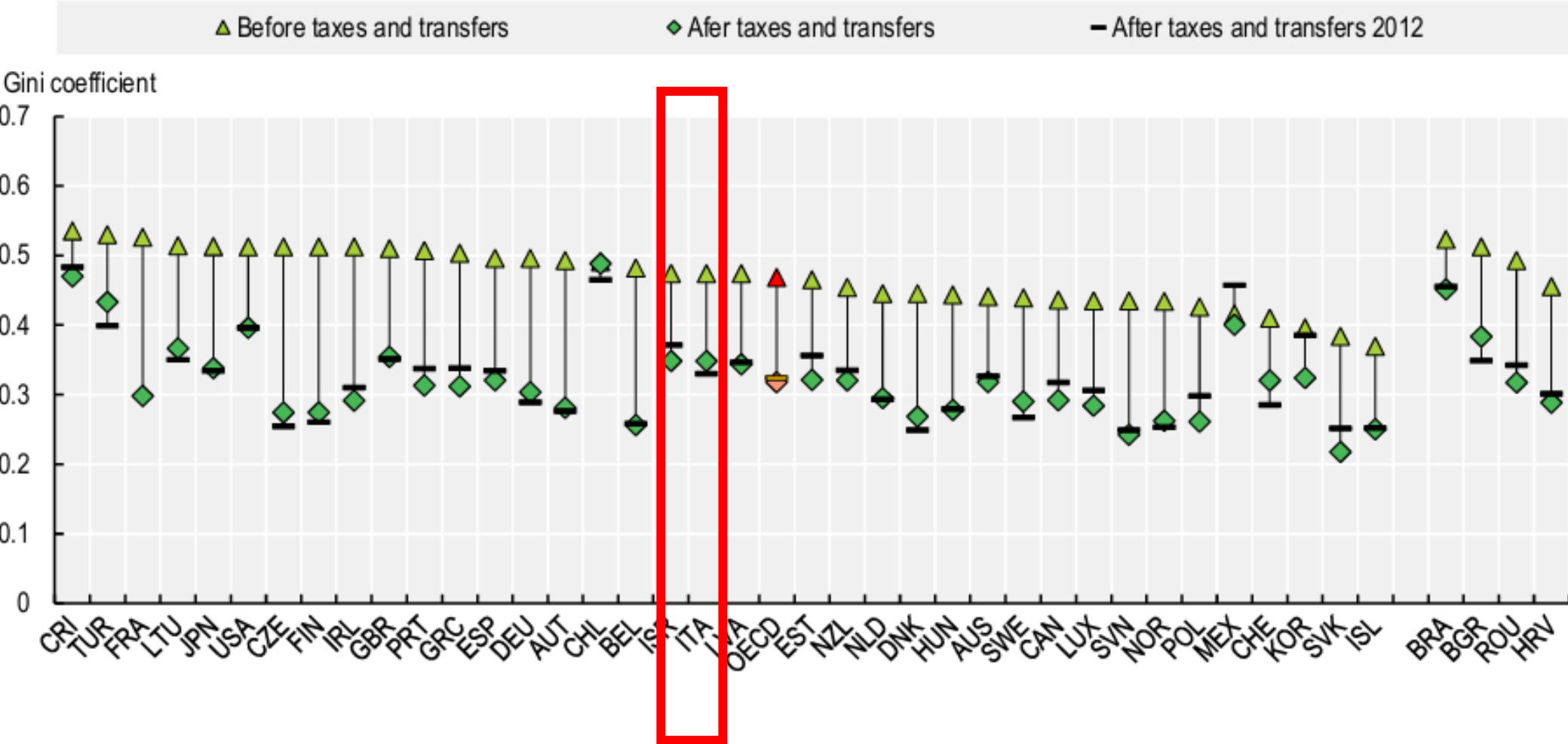


# Relative poverty rates before and after taxes and transfers, 2021 and 2012



Percentage of total population earning less than 50% of median income.  
Source: OECD (2025), *Government at a Glance*

# Household income inequality before and after taxes and transfers, 2021 and 2012



Source: OECD (2025), *Government at a Glance*

## 2. How Should the Government Intervene?

Taxes & subsidies

Regulation

Public provision

Mandates



## 2. How Should the Government Intervene?

- **Tax or Subsidize Private Sale or Purchase**

Use the price mechanism, changing the price of a good to encourage or discourage use

- **Restrict or Mandate Private Sale or Purchase**

Quotas restrict private sale of goods that are overproduced.

Mandates require private purchase of goods that are underproduced.

- **Public Provision**

The government can provide the good directly.

- **Public Financing of Private Provision**

Governments pays; private companies produce.



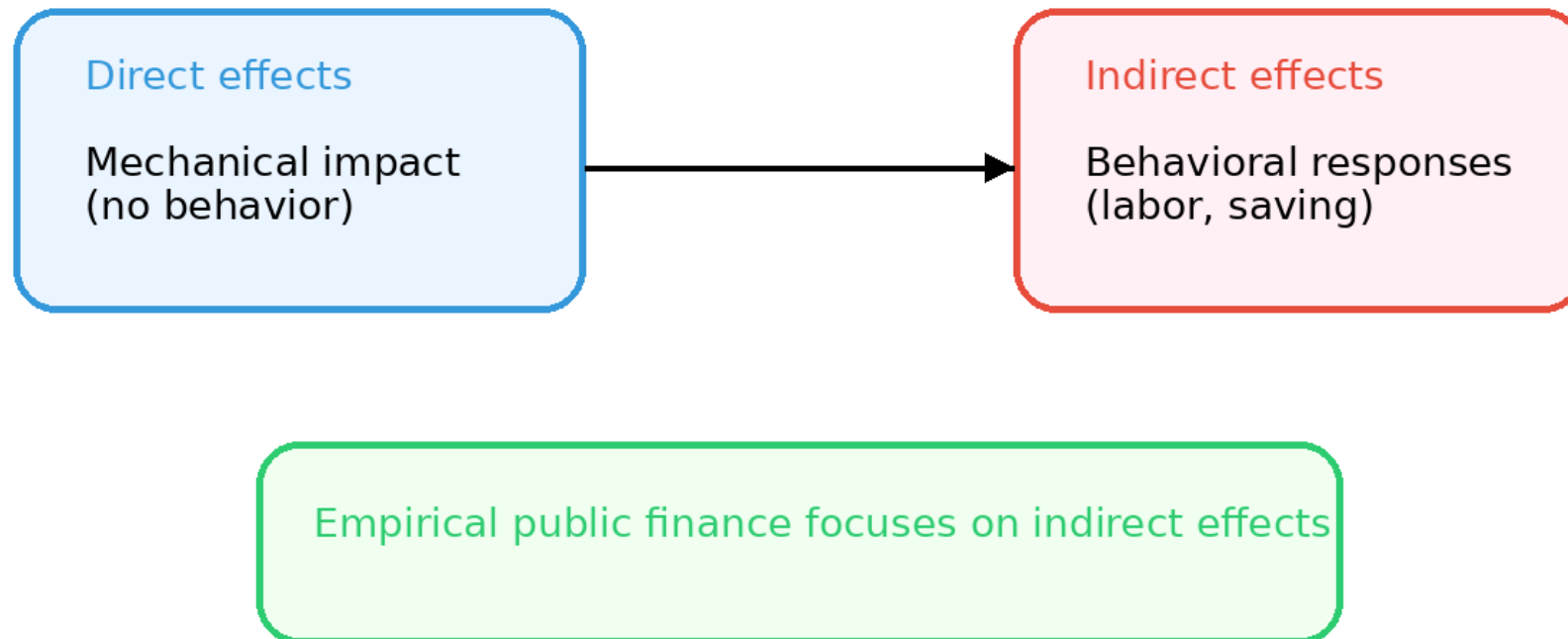
## 2. How Should the Government Intervene?

The regulatory role of the government is directed at a wide range of economic and social activities:

- In the US the Food and Drug Administration (FDA): food, cosmetics, drugs, and medical devices.
- European Food Safety Authority (EFSA) in Parma + European Medicines Agency (EMA) in Amsterdam
- The Environmental Protection Agency (EPA): pollution of air, water, and food supplies.
- European Environment Agency (EEA) in Copenhagen <https://www.eea.europa.eu/>



### 3. What are the Effects of the Intervention?





### 3. What are the Effects of the Intervention?

**Direct effects:** The effects of government interventions that would be predicted if individuals did not change their behavior in response to their interventions.

*With 20 million uninsured, providing universal public health insurance covers 20 million people.*

**Indirect effects:** The effects of government intervention that arise only because individuals change their behavior in response to the interventions.

*If people drop private coverage, many more people may end up covered by the public plan.*



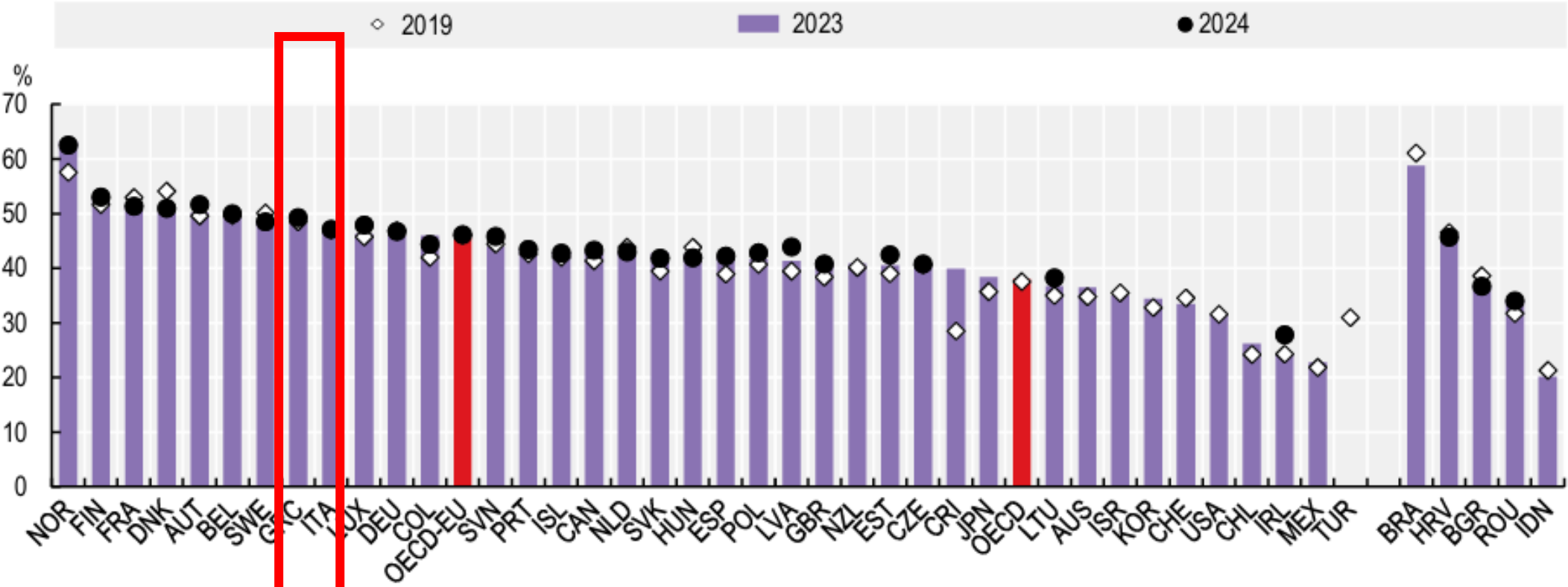
## 4. Why the governments intervene the way they do?

- Governments do not always choose efficient or socially desirable outcomes.
- Governments face enormous challenges in figuring out what the citizens want and how to choose policies that match those wants.

**Political economy: The theory of how the political process produces decisions that affect individuals and the economy**



# General government revenues as a % of GDP, 2019, 2023 and 2024



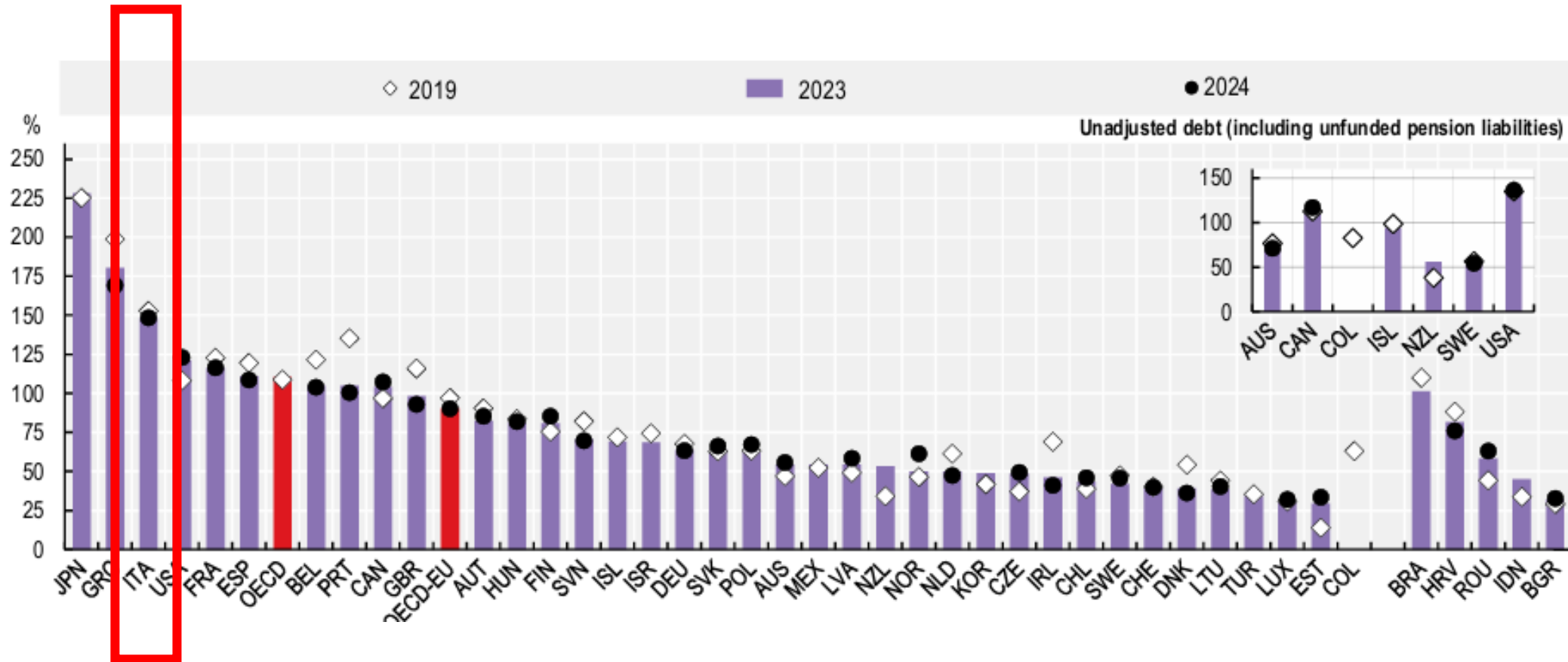
Source: OECD (2025), *Government at a Glance*

# Structure of general government revenues, 2023 and 2024



Source: OECD (2025), *Government at a Glance*

## General government gross debt as a % of GDP, 2019, 2023 and 2024

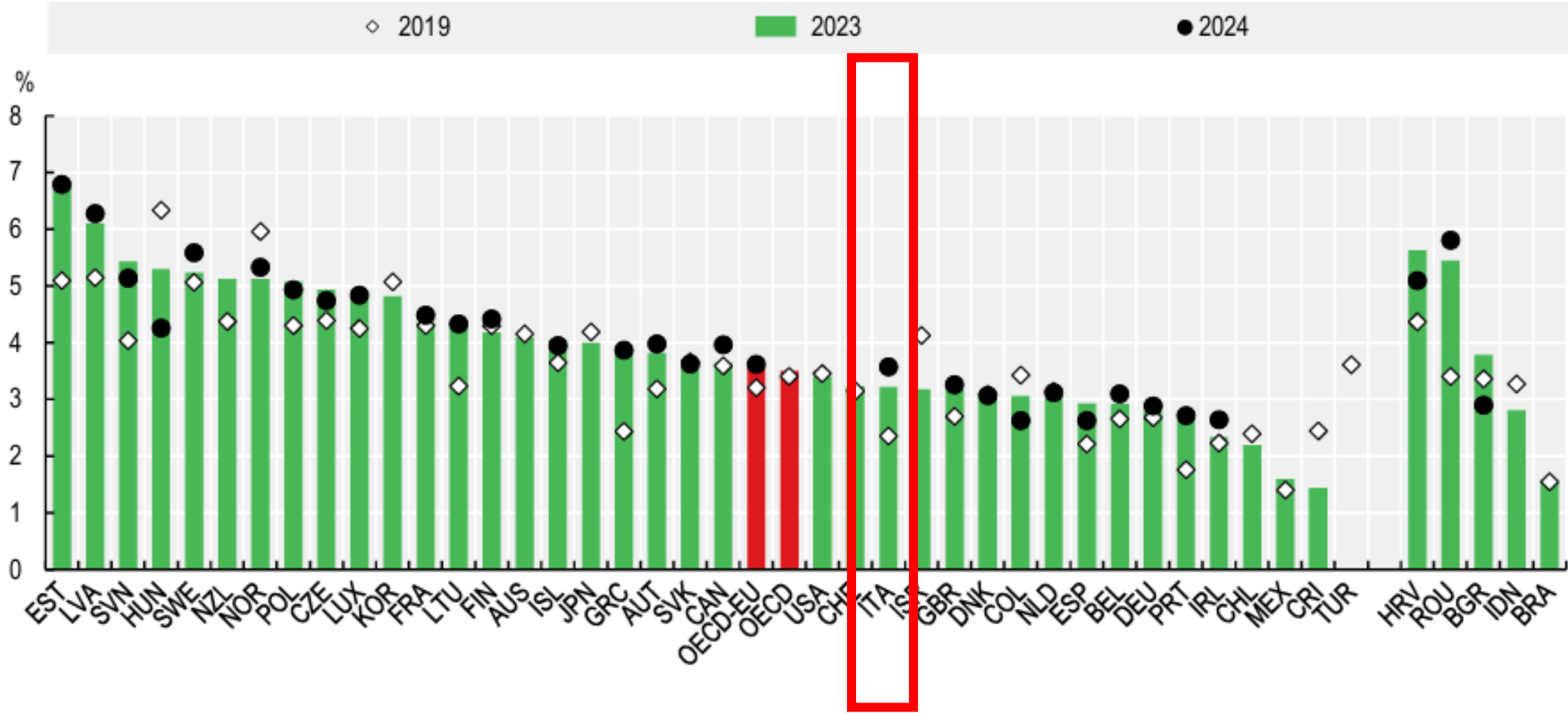


- Deficit = spending – revenues
- Debt = accumulation of past deficits

Source: OECD (2025), *Government at a Glance*



# Government investment as a % of GDP, 2019, 2023 and 2024



Source: OECD (2025), *Government at a Glance*



# Key Takeaways

- Government plays a central role in modern economies.
- A fundamental and ongoing debate concerns the appropriate size and scope of government activity.
- Public finance provides the analytical framework to study the costs, benefits, and consequences of government intervention.

